

Mark Ruangrattham

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EDUCATION

Georgia Institute of Technology

M.S. Computer Science — Computing Systems Track

Atlanta, GA

Dec 2027

University of California, San Diego

B.S. Computer Science

San Diego, CA

Jun 2026

EXPERIENCE

Amazon Web Services, DocumentDB

Software Development Engineer Intern

Seattle, WA

Jun 2026 – Present

- Building a full-stack internal tool adopted org-wide across DocumentDB to streamline the certification process, cutting completion time by 30%.
- Developing an AI-powered failure analysis system that cross-references current and historical test logs, using Amazon's internal AI/MCP tooling to diagnose failing certification tests and recommend fixes, packaged as a custom CLI tool.
- Collaborating with senior SDEs through design and code reviews on production components for a distributed, serverless document database.

Transact Campus + CBORD

Software Engineer Intern

San Diego, CA

Jun – Aug 2025

- Built and deployed an internal developer tooling platform (AWS Lambda, Step Functions, Bedrock) that automated knowledge retrieval and support workflows across 5+ production applications, reducing support ticket volume by 30%.
- Architected a RAG pipeline over AWS Bedrock with embeddings-based document retrieval, enabling scalable knowledge integration across multiple product lines.

UCSD Health

Full Stack Software Engineer

San Diego, CA

Mar 2025 – Jun 2026

- Designed and deployed an internal tooling system (ASP.NET Core, C#, SQL Server, Entity Framework) with RESTful APIs and role-based access control, supporting 150–200 applications/quarter.
- Engineered an admin dashboard with automated workflows and dynamic filtering, reducing manual review time by 60% per quarterly cycle.
- Modernized a legacy application portal using JavaScript, jQuery, and Razor views, saving an estimated \$10K annually in labor hours across hundreds of applicants per cycle.

Valhalla FC

Freelance Software Developer

Remote

Sep 2024 – Present

- Designed and deployed a distributed Python backend from scratch, supporting 200+ concurrent users with 99.9% uptime and sub-second response times via MongoDB-backed real-time data pipelines.
- Engineered a custom Discord bot serving as the core platform for a 2,700+ member community, with a real-time auction engine, tournament bracket management, and automated event workflows.
- Built a custom Discord minigame that increased server interaction by 25%.

PROJECTS

Operating System Kernel | *Java, Virtual Memory, Concurrency, System Design*

- Implemented a systems-level kernel with thread synchronization, virtual memory management (demand paging, TLB, page replacement), and multiprogramming via system calls.
- Built an address translation pipeline with TLB and page table management; engineered a clock algorithm for page replacement and handled race conditions across concurrent processes using locks and condition variables.

CalorieScan | *React, Firebase, Google Gemini*

- Built a full-stack fitness app integrating Google Gemini to track daily calorie and workout intake and surface personalized lifestyle recommendations, with Firebase handling data storage and auth.

SKILLS

Languages: Python, Rust (in progress), C, C++, Java, JavaScript, SQL

Cloud & DB: AWS (Lambda, Bedrock, Step Functions, API Gateway, S3, EC2), GCP, MongoDB, PostgreSQL, SQL Server, Amazon DocumentDB

Frameworks: Django, ASP.NET Core, React, Entity Framework, Pandas, pytest

Tooling & Infra: Git, Docker, Linux/Unix, Bash, asyncio, Postman

Concepts: System Design, Distributed Systems, Concurrency, Database Optimization, RAG/Retrieval Systems, Developer Tooling